US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12399613
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Suzuki; Atsushi

Display device and display method

Abstract

A display device includes: a display unit including a touch panel; and a control unit configured to display, on the display unit, a character input area in which a soft key is arranged, repeatedly receive, from the character input area, an operation for inputting an input character, and display, on the display unit, an input character string display area in which a character string that is a sequence of the input character is displayed. The control unit is configured to arrange in order, in the input character string display area, one of the input character and a replacement character for replacing the input character, acquire from a database including input character confirmation information for confirming a character, when receiving the operation for inputting the input character from the character input area, the input character confirmation information corresponding to the input character, and display the acquired input character confirmation information on the display unit.

Inventors: **Suzuki; Atsushi (Matsumoto, JP)**

Applicant: **SEIKO EPSON CORPORATION** (Tokyo, JP)

Family ID: **1000008781615**

Assignee: **SEIKO EPSON CORPORATION (Tokyo, JP)**

Appl. No.: **17/897385**

Filed: **August 29, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230066745 A1	Mar. 02, 2023

Foreign Application Priority Data

JP	2021-138494	Aug. 27, 2021

Publication Classification

Int. Cl.: **G06F3/04886** (20220101); **G06F3/0481** (20220101); **G06V30/262** (20220101)

U.S. Cl.:

CPC **G06F3/04886** (20130101); **G06F3/0481** (20130101); **G06V30/262** (20220101);

Field of Classification Search

CPC: G06F (3/04886); G06F (3/0481); G06F (21/84); G06F (21/45); G06F (3/016); G06V (30/262); H04L (63/083)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10440007	12/2018	Ramalingam	N/A	G06F 21/84
2006/0217985	12/2005	Noguchi et al.	N/A	N/A
2010/0005525	12/2009	Fischer	726/21	G06F 21/31
2016/0012617	12/2015	Maetz	345/593	G06F 21/31
2017/0223201	12/2016	Ueno	N/A	G06F 3/0233
2019/0250815	12/2018	Tashiro	N/A	G06F 3/0237
2022/0017042	12/2021	Zhang	N/A	B60R 25/23

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2001166843	12/2000	JP	N/A
2006268319	12/2005	JP	N/A
2014142681	12/2013	JP	N/A
2019045946	12/2018	JP	N/A

OTHER PUBLICATIONS

Office Action for JP Patent Application No. 2021138494, issued on Apr. 1, 2025, 6 pages. cited by applicant

---

Primary Examiner: Cruz; Iriana

Attorney, Agent or Firm: CHIP LAW GROUP

Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2021-138494, filed Aug. 27, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.
BACKGROUND
1. Technical Field

(2) The present disclosure relates to a display device, a display method, and a display program for receiving input of a character string.

2. Related Art

(3) When receiving input of a character string such as a password or an SSID, a smart device such as a smartphone or a tablet terminal displays a software keyboard and repeatedly receives an operation for inputting an input character from the software keyboard. Here, the SSID is an abbreviation of service set identifier. The input character is displayed as it is or replaced with a predetermined character such as "*" and arranged in order.

(4) A mobile terminal disclosed in JP-A-2001-166843 temporarily displays a character selected by an operator, and hides the character by displaying an asterisk "*" when a confirmation key is pressed.

(5) It is required in a smart device that a character is input in a small area for display and input, and it is difficult to execute blind touch as in a physical keyboard. Therefore, the user may pay close attention to a software keyboard and may not notice erroneous input.

(6) The problem also exists in a display device other than the smart device.

SUMMARY

(7) A display device according to an aspect of the present disclosure includes: a display unit including a touch panel; and a control unit configured to display, on the display unit, a character input area in which a soft key is arranged, repeatedly receive, from the character input area, an operation for inputting an input character, and display, on the display unit, an input character string display area in which a character string that is a sequence of the input character is displayed. The control unit is configured to arrange in order, in the input character string display area, one of the input character and a replacement character for replacing the input character, acquire from a database including input character confirmation information for confirming a character, when receiving the operation for inputting the input character from the character input area, the input character confirmation information corresponding to the input character, and display the acquired input character confirmation information on the display unit.

(8) A display method according to an aspect of the present disclosure is a display method for displaying, on a display unit including a touch panel, a character input area in which a soft key is arranged; repeatedly receiving, from the character input area, an operation for inputting an input character; displaying, on the display unit, an input character string display area in which a character string that is a sequence of the input character is displayed; and arranging in order, in the input character string display area, one of the input character and a replacement character for replacing the input character. The display method includes: an input character confirmation information acquisition step of acquiring from a database including input character confirmation information for confirming a character, when the operation for inputting the input character is received from the character input area, the input character confirmation information corresponding to the input character; and an input character confirmation information display step of displaying the acquired input character confirmation information on the display unit.

(9) A non-transitory computer-readable storage medium according to an aspect of the present disclosure stores a display program which is a display program for displaying, on a display unit including a touch panel, a character input area in which a soft key is arranged, repeatedly receiving, from the character input area, an operation for inputting an input character, displaying, on the display unit, an input character string display area in which a character string that is a sequence of the input character is displayed, and arranging in order, in the input character string display area, one of the input character and a replacement character for replacing the input character. The display program causes a computer to implement: an input character confirmation information acquisition function of acquiring from a database including input character confirmation information for confirming a character, when the operation for inputting the input character is received from the character input area, the input character confirmation information corresponding to the input character

character; and an input character confirmation information display function of displaying the acquired input character confirmation information on the display unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram schematically showing a configuration example of a system including a display device.

(2) FIG. **2** is a diagram schematically showing an example of a password input screen displayed on the display device.

(3) FIG. **3** is a diagram schematically showing a configuration example of input character confirmation information included in a database.

(4) FIG. **4** is a diagram schematically showing a configuration example of the database including the input character confirmation information.

(5) FIG. **5** is a flowchart showing an example of a group setting process.

(6) FIG. **6** is a diagram schematically showing an example of a confirmation information selection area displayed on a display unit.

(7) FIG. **7** is a diagram schematically showing an example of an editing area displayed on the display unit.

(8) FIG. **8** is a flowchart showing an example of character string input reception process.

(9) FIG. **9**A is a diagram schematically showing another example of an input character string display area; FIG. **9**B is a diagram schematically showing an example of an SSID input screen displayed on the display unit; and FIG. **9**C is a diagram schematically showing a display example of an immediately previous input character display area.

(10) FIG. **10** is a diagram schematically showing an example of an animation display of an input character.

(11) FIG. **11** is a flowchart showing another example of the group setting process.

(12) FIG. **12** is a diagram schematically showing another example of the password input screen displayed on the display device.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(13) Hereinafter, embodiments of the present disclosure will be described. Of course, the following embodiments are merely examples of the present disclosure, and all of features shown in the embodiments are not necessarily essential to the solution to problems of the present disclosure.

1. Outline of Technique Included in Present Disclosure

(14) First, an outline of a technique included in the present disclosure will be described with reference to examples shown in FIGS. **1** to **12**. The drawings of the present application are diagrams schematically showing examples, and enlargement ratios in respective directions shown in these drawings may be different from each other, and the respective drawings may not be consistent with each other. Of course, each element of the present technique is not limited to the specific examples indicated by reference numerals. In the “Outline of Technique Included in Present Disclosure”, words in parentheses mean a supplementary explanation of an immediately previous word.

(15) Aspect 1

(16) As shown in FIG. **1** and the like, a display device **1** according to an aspect of the present technique includes a display unit **15** and a control unit **10**. The display unit **15** includes a touch panel **16**. As shown in FIG. **2** and the like, the control unit **10** displays, on the display unit **15**, a character input area **20** in which soft keys **21** are arranged, repeatedly receives, from the character input area **20**, an operation for inputting an input character **82**, and displays, on the display unit **15**, an input character string display area **30** in which a character string **80** that is a sequence of the

input characters **82** is displayed. As shown in FIGS. **2**, **8**, **9**A, and the like, the control unit **10** arranges in order, in the input character string display area **30**, one of the input character **82** and a replacement character **83** for replacing the input character **82**. As shown in FIGS. **3** and **8**, when receiving, from the character input area **20**, the operation for inputting the input character **82**, the control unit **10** acquires input character confirmation information IN**1** corresponding to the input character **82** from a database DB**1** including input character confirmation information IN**1** for confirming characters **90**. As shown in FIGS. **2**, **8**, and the like, the control unit **10** displays the acquired input character confirmation information IN**1** on the display unit **15**.
(17) When a user performs an operation on the character input area **20**, one of the input character **82** and the replacement character **83** for replacing the input character **82** is arranged in order in the input character string display area **30**. The input character confirmation information IN**1** corresponding to the input character **82** is displayed on the display unit **15**. The user can confirm whether the input character **82** is correct by viewing the input character confirmation information IN**1** corresponding to the input character **82**. As described above, in Aspect 1, since what an immediately previous input character is can be easily confirmed, erroneous input can be reduced. For example, when the character string is important registration information, it is easy to accurately input the important registration information, and failure of setting can be reduced.
(18) Here, a key layout in the character input area **20** is not limited to a standardized keyboard layout, and may be a non-standardized layout such as in alphabetical order or the like.
(19) The database DB**1** may be stored in the display device itself or may be stored in an external computer such as a server **300**.
(20) The additional statements are also applied to the following aspects.
(21) Aspect 2
(22) As shown in FIGS. **2** and **9**B, the character string **80** may be one of a password and an identifier. As shown in FIG. **3** and the like, the input character confirmation information IN**1** may include at least one of a phonetic code of the character **90**, a word having the character **90** as an initial character, and a pronunciation of the character **90**. In the present aspect, it is possible to easily input the password or the identifier that is the important registration information.
(23) Here, the identifier includes an SSID, an IP address, a domain name, a host name, a port number, a MAC address, a mail address, and the like. The IP address is an abbreviation of internet protocol address, and the MAC address is an abbreviation of media access control address.
(24) The phonetic code is also referred to as a spelling alphabet, and is a word used in an international unified manner.
(25) The additional statements are also applied to the following aspects.
(26) Aspect 3
(27) As shown in FIG. **2** and the like, the control unit **10** may display, on the display unit **15**, an immediately previous input character display area **40**, in which an immediately previous input character **84** input in response to immediately previous reception of the operation is displayed, separately from the input character string display area **30**. In the aspect, since the immediately previous input character **84** input in response to the immediately previous reception of the operation is displayed in the immediately previous input character display area **40** separately from the character string **80** displayed in the input character string display area **30**, the immediately previous input character can be more easily confirmed.
(28) Although not included in Aspect 3, an example in which the immediately previous input character display area **40** is not displayed on the display unit **15** as shown in FIG. **12** is also included in the present technique.
(29) Aspect 4
(30) As shown in FIG. **2** and the like, the control unit **10** may display the input character **82** in the immediately previous input character display area **40** in a manner of being larger than a display character **81** in the input character string display area **30**. In this aspect, since the immediately

previous input character **84** input in response to the immediately previous reception of the operation is displayed in a large size, the immediately previous input character can be more easily confirmed.

(31) Aspect 5

(32) As shown in FIG. **2** and the like, the control unit **10** may display the input character confirmation information IN**1** in parentheses in the immediately previous input character display area **40**. In the aspect, since the input character confirmation information IN**1** is displayed in the parentheses according to the immediately previous input character **84** input in response to the immediately previous reception of the operation, the immediately previous input character can be more easily confirmed.

(33) Aspect 6

(34) As shown in FIG. **10**, the control unit **10** may display, on the display unit **15**, the input character **82** input in response to reception of the operation, such that the input character **82** floats from the character input area **20** and moves to the immediately previous input character display area **40**. In the aspect, an animation display **25** is executed in which the input character **82** floats from the character input area **20** and moves to the immediately previous input character display area **40**, and thus the immediately previous input character can be more easily confirmed.

(35) Aspect 7

(36) As shown in FIGS. **2**, **9**B, and **9**C, the control unit **10** may display the input character confirmation information IN**1** on the display unit **15** such that at least one of a typeface and a character size is changed depending on whether the input character **82** is an uppercase character of the alphabet or is a lowercase character of the alphabet. In the aspect, since the input character confirmation information IN**1** is displayed such that at least one of the typeface and the character size when the input character **82** is the uppercase character of the alphabet is different from that when the input character **82** is the lowercase character of the alphabet, the immediately previous input character can be more easily confirmed.

(37) Aspect 8

(38) As shown in FIG. **7**, the control unit **10** may display, on the display unit **15**, an editing area **50** for receiving editing of the input character confirmation information IN**1** to be displayed on the display unit **15**. When receiving the editing of the input character confirmation information IN**1** from the editing area **50**, the control unit **10** may display, on the display unit **15**, the input character confirmation information IN**1** that corresponds to the input character **82** and is obtained in response to reception of the editing. In the present aspect, since the display of the input character confirmation information IN**1** can be customized, the immediately previous input character can be more easily confirmed.

(39) Aspect 9

(40) As shown in FIG. **3** and the like, the input character confirmation information IN**1** may be divided into a plurality of groups of confirmation information IN**2**. As shown in FIGS. **5** and **6**, the control unit **10** may display, on the display unit **15**, a confirmation information selection area **60** for receiving selection of the confirmation information IN**2** to be displayed on the display unit **15** from the plurality of groups of confirmation information IN**2**. When receiving the selection of the confirmation information IN**2** from the confirmation information selection area **60**, the control unit **10** may acquire, from the database DB**1**, the confirmation information IN**2** selected among the input character confirmation information IN**1** corresponding to the input character **82**. The control unit **10** may display the acquired confirmation information IN**2** on the display unit **15**. In the present aspect, since the input character confirmation information IN**1** to be displayed can be selected from the plurality of divided groups of confirmation information IN**2**, the immediately previous input character can be more easily confirmed.

(41) Aspect 10

(42) As shown in FIG. **4**, the input character confirmation information IN**1** may be divided into a

plurality of groups of country-specific or region-specific information IN**3** according to countries or regions. As shown in FIGS. **11** and **8**, the control unit **10** may acquire, from the database DB**1**, the country-specific or region-specific information IN**3** corresponding to a country or a region where the display device **1** is present among the input character confirmation information IN**1** corresponding to the input character **82**. The control unit **10** may display the acquired country-specific or region-specific information IN**3** on the display unit **15**. In the present aspect, since the country-specific or region-specific information IN**3** corresponding to the country or the region where the display device **1** is present among the input character confirmation information IN**1** is displayed, the immediately previous input character can be more easily confirmed.
(43) Aspect 11
(44) As shown in FIG. **4**, the input character confirmation information IN**1** may be divided into a plurality of groups of language-specific information IN**4** according to languages. As shown in FIG. **5**, the control unit **10** may acquire, from the database DB**1**, the language-specific information IN**4** corresponding to a language set in the display device **1** among the input character confirmation information IN**1** corresponding to the input character **82**. The control unit **10** may display the acquired language-specific information IN**4** on the display unit **15**. In the present aspect, since the language-specific information IN**4** corresponding to the language set in the display device **1** among the input character confirmation information IN**1** is displayed, the immediately previous input character can be more easily confirmed.
(45) Aspect 12
(46) As shown in FIGS. **1** and **8**, a display method according to an aspect of the present technique is a display method for displaying, on the display unit **15** including the touch panel **16**, the character input area **20** in which the soft keys **21** are arranged; repeatedly receiving an operation for inputting the input character **82** from the character input area **20**; displaying, on the display unit **15**, the input character string display area **30** in which the character string **80** that is the sequence of the input character **82** is displayed; and arranging in order, in the input character string display area **30**, one of the input character **82** and the replacement character **83** for replacing the input character **82**. The display method includes steps (A) and (B).
(47) Step (A) is an input character confirmation information acquisition step ST**3** of acquiring, from the database DB**1** including the input character confirmation information IN**1** for confirming the character **90**, the input character confirmation information IN**1** corresponding to the input character **82** when the operation for inputting the input character **82** is received from the character input area **20**.
(48) Step (B) is an input character confirmation information display step ST**4** of displaying the acquired input character confirmation information IN**1** on the display unit **15**.
(49) In Aspect 12, since what the immediately previous input character is can also be easily confirmed, the erroneous input can be reduced.
(50) Aspect 13
(51) As shown in FIGS. **1** and **8**, a display program PR**0** according to an aspect of the present technique is a display program for displaying, on the display unit **15** including the touch panel **16**, the character input area **20** in which the soft keys **21** are arranged, repeatedly receiving, from the character input area **20**, the operation for inputting the input character **82**, displaying, on the display unit **15**, the input character string display area **30** in which the character string **80** that is the sequence of the input character **82** is displayed, and arranging in order, in the input character string display area **30**, one of the input character **82** and the replacement character **83** for replacing the input character **82**. The display program PR**0** causes the computer, for example, the display device **1** to implement an input character confirmation information acquisition function FU**3** and an input character confirmation information display function FU**4**. The input character confirmation information acquisition function FU**3** is to acquire, from the database DB**1**, the input character confirmation information IN**1** corresponding to the input character **82** when the operation for

inputting the input character **82** is received from the character input area **20**. The database DB**1** includes the input character confirmation information IN**1** for confirming the character **90**. The input character confirmation information display function FU**4** is to display the acquired input character confirmation information IN**1** on the display unit **15**.
(52) In Aspect 13, since what the immediately previous input character is can also be easily confirmed, the erroneous input can be reduced.
(53) Further, the present technique can be applied to a multifunction device including the above display device, a display method for the multifunction device, a control program for the multifunction device, a computer-readable medium in which the control program or the display program is recorded, and the like. Any one of the devices may include a plurality of distributed portions.
2. Specific Example of Configuration of System Including Display Device
(54) FIG. **1** schematically shows a configuration of a system SY**1** including the display device **1**.
(55) The system SY**1** shown in FIG. **1** includes a wireless router **100**, the display device **1**, a printer **200**, and the server **300**. The display device **1** and the printer **200** are wirelessly coupled to the wireless router **100**. The wireless router **100** is coupled to the server **300** via a network NE**1**. As the network NE**1**, the Internet, a LAN, a combination thereof, or the like can be applied. Here, the LAN is an abbreviation of local area network.
(56) The display device **1** is a computer including the control unit **10**, a storage unit **14**, the display unit **15** including the touch panel **16**, a communication unit **17**, an audio output unit **18**, an audio input unit **19**, and the like. The control unit **10** includes a CPU **11** that is a processor, a ROM **12** that is a semiconductor memory, a RAM **13** that is a semiconductor memory, and the like. Here, the CPU is an abbreviation of central processing unit. The ROM is an abbreviation of read only memory. The RAM is an abbreviation of random access memory. The plurality of elements indicated by the reference numerals **11** to **19** can input and output information by being electrically coupled.
(57) The storage unit **14** stores an OS and the display program PR**0**, and may store the database DB**1** including the input character confirmation information IN**1** shown in FIG. **3**. Here, the OS is an abbreviation of operating system. As the storage unit **14**, a nonvolatile semiconductor memory such as a flash memory, a magnetic storage device such as a hard disk, or the like can be used.
(58) The display program PR**0** is an application program that causes the display device **1** to implement a plurality of functions shown in FIG. **8**, for example, an input character string display function FU**1**, an immediately previous input character display function FU**2**, the input character confirmation information acquisition function FU**3**, and the input character confirmation information display function FU**4**. The plurality of functions indicated by the reference numerals FU**1** to FU**4** in FIG. **8** are exhibited by installing the display program PR**0** in the display device **1**, reading the display program PR**0** into the RAM **13**, and executing the display program PR**0** by the CPU **11**. As a result, a plurality of steps as shown in FIG. **8**, for example, an input character string display step ST**1** corresponding to the input character string display function FU**1**, an immediately previous input character display step ST**2** corresponding to the immediately previous input character display function FU**2**, the input character confirmation information acquisition step ST**3** corresponding to the input character confirmation information acquisition function FU**3**, and the input character confirmation information display step ST**4** corresponding to the input character confirmation information display function FU**4** are executed.
(59) As the display unit **15**, a liquid crystal display panel or the like can be used. The touch panel **16** is attached to a surface of the display unit **15** and receives a touch operation executed by the user. As the touch panel **16**, a capacitive touch panel that senses a change in capacitance of the surface due to touch, or the like can be used. The touch panel **16** may be a multi-touch type touch panel capable of detecting touch at a plurality of portions at the same time, or may be a single-touch type touch panel capable of detecting touch at only one portion.

(60) The communication unit **17** is an interface that wirelessly communicates with a communication unit **106** of the wireless router **100** according to a wireless LAN standard such as Wi-Fi (registered trademark). The communication unit **17** can transmit print data to the printer **200** or receive a status from the printer **200** via the wireless router **100**. The audio output unit **18** generates audio according to an audio signal. As the audio output unit **18**, a general-purpose speaker can be used. The audio input unit **19** converts input audio into an audio signal. As the audio input unit **19**, a general-purpose microphone can be used. When the display device **1** is a mobile phone such as a smartphone, during a phone call, the audio output unit **18** generates audio according to an audio signal from a telephone line, and the audio input unit **19** converts audio from a user or the like into an audio signal to be output to the telephone line.
(61) The display device **1** to which the present technique can be applied includes a smart device such as a smartphone or a tablet terminal, a personal computer including a display, a printer including a display panel, an ATM, that is, an automatic teller machine, and the like. It is assumed that the display device **1** in the specific example is a smart device in which the display unit **15** is relatively small.
(62) The wireless router **100** includes a controller **101**, a communication I/F **105**, the communication unit **106**, and the like, and functions as an access point. Here, I/F is an abbreviation of interface. The controller **101** includes a CPU **102**, a ROM **103**, a RAM **104**, and the like, and stores the SSID that is a unique identification name, the password, and the like. The communication I/F **105** is an interface that communicates with the server **300** via the network NE**1**. The communication unit **106** is an interface that wirelessly communicates with the communication unit **17** of the display device **1** and a communication unit **206** of the printer **200** according to the wireless LAN standard such as the Wi-Fi (registered trademark). The communication unit **106** confirms the SSID and the password from the communication units **17** and **206**, and then wirelessly communicates with the communication units **17** and **206**.
(63) The printer **200** includes a CPU **201**, a ROM **202**, a RAM **203**, a storage device **204**, a printing unit **205**, the communication unit **206**, and the like. As the storage device **204**, a nonvolatile semiconductor memory, a magnetic storage device, or the like can be used. The printing unit **205** executes printing based on the print data received from the display device **1** via the wireless router **100**. As the printing unit **205**, an inkjet type print engine that ejects an ink droplet onto a print medium, an electrophotographic print engine that attaches toner to the print medium, or the like can be used. The communication unit **206** is an interface that wirelessly communicates with the communication unit **106** of the wireless router **100** according to the wireless LAN standard such as the Wi-Fi (registered trademark). The communication unit **206** can transmit the print data from the display device **1** or transmit the status to the display device **1** via the wireless router **100**.
(64) The printer **200** may be a copier, a facsimile machine, a multifunction peripheral having a document reading function and a document printing function, or the like.
(65) The server **300** is a computer including a CPU **301**, a ROM **302**, a RAM **303**, a storage device **304**, a communication I/F **305**, and the like. The storage device **304** may store the database DB**1** including the input character confirmation information IN**1** shown in FIG. **3**. The communication I/F **305** is an interface that communicates with the wireless router **100** via the network NE**1**.
(66) In order to cause the printer **200** to execute the printing by operating the display device **1** in the system SY**1** described above, first, it is necessary to wirelessly connect the display device **1** to the wireless router **100** by setting the SSID and the password in the display device **1**. Regarding the SSID, the display device **1** may automatically acquire the SSID based on an SSID acquisition function that is implemented by the OS on the display device **1**, but it is necessary to input at least the password to the display device **1** by the user. When the password or the like is not accurately input to the display device **1**, the wireless connection of the display device **1** to the wireless router **100** ends in failure.
(67) FIG. **2** schematically shows a password input screen displayed on the display unit **15** of the

display device **1**. When the control unit **10** of the display device **1** receives, by the touch panel **16**, an operation for wirelessly connecting the display device **1** to the wireless router **100**, the control unit **10** displays the password input screen shown in FIG. **2** on the display unit **15**. The password input screen includes the character input area **20**, the input character string display area **30**, the immediately previous input character display area **40**, a keyboard switching button **71**, an OK button **72**, and a cancel button **73**.

(68) In the character input area **20**, the plurality of soft keys **21** that are used to receive a tap operation are arranged in an orderly manner. The arrangement of the soft keys **21** is a kind of software keyboard. Here, functions of the OS include a function of displaying, on the display unit **15**, a software keyboard having a QWERTY layout that is a de facto standard of a key layout. However, since the software keyboard having the QWERTY layout is displayed on the relatively small display unit **15**, it is necessary to operate a small soft key, and it is also difficult to execute blind touch on the OS standard software keyboard like a physical keyboard. For this reason, the user often pays close attention to the software keyboard and does not notice the erroneous input of the character. Therefore, in this specific example, the control unit **10** displays, in the character input area **20**, the plurality of soft keys **21** in which numbers, lowercase characters of the alphabet, uppercase characters of the alphabet, symbols, and the like are classified and ordered.

(69) The plurality of soft keys **21** shown in FIG. **2** include a plurality of number keys **21***a*, a plurality of lowercase character keys **21***b*, a plurality of uppercase character keys **21***c*, a plurality of symbol keys **21***d*, a backspace key **21***e*, and an enter key **21***f*. These soft keys **21** are color-coded according to types. Character keys to be operated to input the input character **82** include the plurality of number keys **21***a*, the plurality of lowercase character keys **21***b*, the plurality of uppercase character keys **21***c*, and the plurality of symbol keys **21***d*. Ten kinds of numbers from 1 to 0 are arranged in order on the plurality of number keys **21***a*. 26 kinds of lowercase characters are arranged in alphabetical order on the plurality of lowercase character keys **21***b*. 26 kinds of uppercase characters are arranged in alphabetical order on the plurality of uppercase character keys **21***c*. Therefore, the user can easily search for numbers, lowercase characters of the alphabet, and uppercase character keys of the alphabet, and can easily operate the number keys **21***a*, the lowercase character keys **21***b*, and the uppercase character keys **21***c*. When receiving an operation on any one of the number keys **21***a*, the lowercase character keys **21***b*, the uppercase character keys **21***c*, and the symbol keys **21***d* by the touch panel **16**, the control unit **10** adds the input character **82** to a tail of the character string **80** in the input character string display area **30**. When receiving an operation on the backspace key **21***e*, the control unit **10** deletes a character at the tail of the character string **80**. The enter key **21***f* is a soft key that is used to confirm the character string **80** displayed in the input character string display area **30**.

(70) The control unit **10** repeatedly receives, from the character input area **20**, the tap operation for inputting the input character **82**.

(71) The character string **80** that is the sequence of the input characters **82** is displayed in the input character string display area **30**. Every time receiving, from the character input area **20**, the operation for inputting the input character **82**, the control unit **10** arranges the input character **82** in the input character string display area **30** in order from a left end. For example, when receiving an operation for inputting "A", the control unit **10** adds the "A" that is the immediately previous input character **84** to a right end that is the tail of the character string **80** in the input character string display area **30**. The display character **81** in the input character string display area **30** shown in FIG. **2** is the input character **82** itself.

(72) However, the input characters **82** are arranged in the input character string display area **30** displayed on the relatively small display unit **15**, and thus each of the input characters **82** is small. Therefore, the user often does not notice the erroneous input of the character. Therefore, in this specific example, the control unit **10** displays, on the display unit **15**, the input character confirmation information IN**1** for confirming the immediately previous input character **84**, and the

immediately previous input character display area **40** in which the immediately previous input character **84** is separately displayed.
(73) In the immediately previous input character display area **40**, the input character input in response to the immediately previous reception of the operation, that is, the immediately previous input character **84** is displayed in an enlarged manner, and the input character confirmation information IN**1** for confirming the immediately previous input character **84** is displayed in the parentheses. The control unit **10** displays the immediately previous input character **84** in the immediately previous input character display area **40** in a manner of being larger than the display character **81** in the input character string display area **30**, and larger than a character in the character input area **20**. The control unit **10** displays the input character confirmation information IN**1** in the immediately previous input character display area **40** with the same character size as that of the immediately previous input character **84**. Since the immediately previous input character **84** and the input character confirmation information IN**1** are displayed in the immediately previous input character display area **40** that is different from the input character string display area **30**, the immediately previous input character **84** and the input character confirmation information IN**1** are easily understood, and the user can easily confirm the immediately previous input character. Since the immediately previous input character **84** and the input character confirmation information IN**1** are displayed in the large size, the immediately previous input character **84** and the input character confirmation information IN**1** are easily understood, and the user can more easily confirm the immediately previous input character. Although details will be described later, the input character confirmation information IN**1** for confirming the immediately previous input character **84** is acquired from the database DB**1**, and is displayed adjacent to the immediately previous input character **84** in the immediately previous input character display area **40**.
(74) When receiving an operation on the keyboard switching button **71**, the control unit **10** displays the software keyboard having the QWERTY layout on the display unit **15**. When receiving an operation on the OK button **72**, the control unit **10** determines the character string **80** displayed in the input character string display area **30**, and transmits the character string **80** to a next process. When receiving an operation on the cancel button **73**, the control unit **10** discards the character string **80** displayed in the input character string display area **30** and returns to a process before password input.
(75) FIG. **3** schematically shows a configuration of the input character confirmation information IN**1** included in the database DB**1** that is stored in the server **300** or the display device **1**.
(76) The input character confirmation information IN**1** is information for confirming the character **90**. The input character confirmation information IN**1** shown in FIG. **3** is divided into a plurality of groups of confirmation information IN**2**. The plurality of groups of confirmation information IN**2** includes confirmation information IN**2***a* with emphasis on the phonetic code, confirmation information IN**2***b* with emphasis on an easy-to-understand word, and confirmation information IN**2***c* with emphasis on the pronunciation. Each confirmation information IN**2** includes information associated with the character **90**.
(77) Regarding the phonetic code, the word having the character **90** as the initial character regarding the alphabet is adopted, and the phonetic code is used uniformly in many industries. In the confirmation information IN**2***a* with emphasis on the phonetic code, for example, a phonetic code “ALPHA” is associated with an uppercase character “A”, and a phonetic code “BRAVO” is associated with an uppercase character “B”. The phonetic code such as “alpha” is associated, in a lowercase character form, with the lowercase character. Regarding the number, a word indicating the pronunciation of the character **90** such as “One” is associated with the number.
(78) However, the phonetic code may be a word that is not familiar to the user. Therefore, the input character confirmation information IN**1** includes the confirmation information IN**2***b* including the easy-to-understand word which has the character **90** as the initial character of the alphabet. In the confirmation information IN**2***b*, for example, the well-known “APPLE” is associated with a

character "A", and the well-known "BANANA" is associated with a character "B". The easy-to-understand word such as "apple" is associated, in a lowercase character form, with the lowercase character. Regarding the symbol, the easy-to-understand word is also associated with the symbol, for example, "Underline" instead of "Underscore" is associated with "_". Regarding the number, since it is easy to understand the number even by the phonetic code, the same word as the phonetic code is associated with the number.
(79) In the confirmation information IN**2**c with emphasis on the pronunciation, regarding the alphabet, a phonetic symbol of the character **90** such as "ei" is associated with the character **90**. The phonetic symbol is an example of the pronunciation. Here, a phonetic symbol whose typeface is larger than that of a phonetic symbol associated with the lowercase character is associated with the uppercase character. Regarding the number, since the pronunciation is also indicated by the phonetic code, the same word as the phonetic code is associated with the number.
(80) As described above, the input character confirmation information IN**1** includes the phonetic code of the character **90** such as "ALPHA", the word having the character **90** as the initial character such as "APPLE", and the pronunciation of the character **90** such as "ei" or "One".
(81) FIG. **4** schematically shows a configuration of the database DB**1** including the input character confirmation information IN**1** described above. As shown in FIG. **4**, the input character confirmation information IN**1** is divided into a plurality of groups of country-specific or region-specific information IN**3** according to countries or regions, and is divided into a plurality of groups of language-specific information IN**4** according to languages.
(82) The plurality of groups of country-specific or region-specific information IN**3** shown in FIG. **4** includes country A confirmation information IN**3**a, country B confirmation information IN**3**b, country C confirmation information IN**3**c, region D confirmation information IN**3**d, and the like. The plurality of groups of language-specific information IN**4** shown in FIG. **4** includes language a confirmation information IN**4**a, language b confirmation information IN**4**b, language e confirmation information IN**4**c, language c confirmation information IN**4**d, and the like. In FIG. **4**, a relation between the plurality of groups of country-specific or region-specific information IN**3** and the plurality of groups of language-specific information IN**4** is indicated by a two-dot chain line arrow.
(83) For example, when one type of language is used in the country A and the region D, the language-specific information IN**4** corresponding to the used language can be used as the country-specific or region-specific information IN**3**. In the example shown in FIG. **4**, the language a confirmation information IN**4**a is used in the country A confirmation information IN**3**a, and the language c confirmation information IN**4**d is used in the region D confirmation information IN**3**d. The country B confirmation information IN**3**b for country B where languages a, b, and e are used includes the language a confirmation information IN**4**a, language b confirmation information IN**4**b, and language e confirmation information IN**4**c. Therefore, it can be said that the country B confirmation information IN**3**b is divided into the language a confirmation information IN**4**a, the language b confirmation information IN**4**b, and the language e confirmation information IN**4**c.
(84) Hereinafter, a process example of setting a group for displaying the input character confirmation information IN**1** according to language setting of the display device **1** will be described with reference to FIG. **5** and the like.
3. Specific Example of Process Executed by Display Device
(85) FIG. **5** shows a group setting process executed by the control unit **10** of the display device **1**. This process includes processes of steps S**102** to S**108**. Hereinafter, the description of "step" is omitted, and reference numerals of the steps are shown in parentheses. The group setting process is executed, for example, when the display program PR**0** is installed in the display device **1**.
(86) When the group setting process is started, the control unit **10** acquires the language setting indicating a language set in the display device **1** (S**102**). The OS of the display device **1** causes the display device **1** to implement a function of receiving the language setting and storing the language

setting in the storage unit **14**, and a function of reading the language setting from the storage unit **14** in response to a language setting acquisition request from an application program and transmitting the language setting to the application program. Therefore, the control unit **10** can execute a process of acquiring the language setting from the OS by issuing the language setting acquisition request to the OS. After acquiring the language setting, the control unit **10** executes a process of setting the language-specific information IN**4** corresponding to the acquired language setting as the input character confirmation information IN**1** to be used, for example, a process of storing, in the storage unit **14**, setting information indicating the language-specific information IN**4** corresponding to the language setting (S**104**).

(87) The input character confirmation information IN**1**, which is the language-specific information IN**4** corresponding to the language setting, may be divided into a plurality of groups of confirmation information IN**2** as shown in FIG. **3**. When the input character confirmation information IN**1** is divided into groups, the control unit **10** displays the confirmation information selection area **60** as shown in FIG. **6** on the display unit **15** (S**106**).

(88) FIG. **6** schematically shows the confirmation information selection area **60** displayed on the display unit **15** based on the input character confirmation information IN**1** shown in FIG. **3**. The confirmation information selection area **60** is an area for receiving selection of confirmation information to be displayed on the display unit **15** from the plurality of groups of confirmation information IN**2**. The confirmation information selection area **60** shown in FIG. **6** includes selection areas **60***a*, **60***b*, and **60***c* for selecting the confirmation information IN**2**, an editing button **60***e*, a setting end button **60***f*, and a cancel button **60***g*. The selection area **60***a* is an area for selecting the confirmation information IN**2***a* with emphasis on the phonetic code. The selection area **60***b* is an area for selecting the confirmation information IN**2***b* with emphasis on the easy-to-understand word. The selection area **60***c* is an area for selecting the confirmation information IN**2***c* with emphasis on the pronunciation. FIG. **6** shows that a radio button of the selection area **60***b* is turned on by executing an operation of selecting the selection area **60***b* corresponding to the confirmation information IN**2***b* with emphasis on the easy-to-understand word. When receiving an operation on the setting end button **60***f*, the control unit **10** stores, in the storage unit **14**, group setting indicating confirmation information corresponding to an area selected from the selection areas **60***a*, **60***b*, and **60***c* (S**108** of FIG. **5**), and ends the group setting process. When receiving an operation on the cancel button **60***g*, the control unit **10** stores default group setting in the storage unit **14** and ends the group setting process.

(89) Depending on the user, it may be desired to customize a part of the input character confirmation information IN**1** while following the group setting as a whole. Therefore, the editing button **60***e* for editing a part of the input character confirmation information IN**1** is provided in the confirmation information selection area **60** shown in FIG. **6**. When receiving an operation on the editing button **60***e*, the control unit **10** displays the editing area **50** on the display unit **15** as shown in FIG. **7**.

(90) FIG. **7** schematically shows the editing area **50** displayed on the display unit **15**. The editing area **50** is an area for receiving the editing of the input character confirmation information IN**1** to be displayed on the display unit **15**. The editing area **50** shown in FIG. **7** includes an editing field **50***a*, an editing end button **50***b*, and a cancel button **50***c*. In the editing field **50***a*, a list of the input character confirmation information IN**1** associated with the character **90** is shown. The input character confirmation information IN**1** in the editing field **50***a* shown in FIG. **7** is one group of confirmation information selected from the plurality of groups of confirmation information IN**2**. The control unit **10** receives, by the touch panel **16**, an operation for editing the input character confirmation information IN**1** in the editing field **50***a*. For example, when the user executes, using a software keyboard (not shown), an operation for changing the input character confirmation information “BANANA” associated with the uppercase character “B” to “BONUS”, the control unit **10** associates the input character confirmation information “BONUS” with the uppercase

character “B”. At this time, the control unit **10** may change “banana” associated with the lowercase character “b” that is the same alphabet as the uppercase character “B” to “bonus”. When receiving an operation on the editing end button **50***b*, the control unit **10** stores the edited input character confirmation information IN**1** in, for example, the database DB**1**, and deletes the editing area **50** from the display unit **15**. An edited content may be stored in the storage unit **14** separately from the database DB**1**. When receiving an operation on the cancel button **50***c*, the control unit **10** discards the edited content and deletes the editing area **50** from the display unit **15**.

(91) FIG. **8** shows a character string input reception process executed by the control unit **10**. Here, S**206** corresponds to the input character string display step ST**1** and the input character string display function FU**1**. S**208** corresponds to the immediately previous input character display step ST**2** and the immediately previous input character display function FU**2**. S**210** corresponds to the input character confirmation information acquisition step ST**3** and the input character confirmation information acquisition function FU**3**. S**212** corresponds to the input character confirmation information display step ST**4** and the input character confirmation information display function FU**4**. The character string input reception process is executed, for example, when the operation for wirelessly connecting the display device **1** to the wireless router **100** is received by the touch panel **16**.

(92) When the character string input reception process is started, the control unit **10** displays an initial screen on the display unit **15** (S**202**). Although not shown, the initial screen is a screen in a state in which no input character **82** is displayed in the input character string display area **30** and the immediately previous input character display area **40** on the password input screen shown in FIG. **2**.

(93) After the control unit **10** displays the initial screen, the process branches according to the operation on the character input area **20** or the like (S**204**).

(94) When the character keys indicated by the reference numerals **21***a*, **21***b*, **21***c*, and **21***d* are operated, the control unit **10** arranges the input characters **82** corresponding to the operated character keys in order in the input character string display area **30** (S**206**). Then, as in the input character string display area **30** shown in FIG. **2**, the character string **80** that is the sequence of the input characters **82** is displayed in the input character string display area **30**.

(95) Here, as shown in FIG. **9**A, the replacement character **83** for replacing the input character **82** may be arranged in order in the input character string display area **30**. FIG. **9**A schematically shows another example of the input character string display area **30**.

(96) In the input character string display area **30**, the control unit **10** arranges “*” that is the replacement character **83** for replacing the input character **82**, except for the immediately previous input character **84**. Since the replacement character **83** only hides the input character **82**, the character string **80** shown in the input character string display area **30** can be said to be the sequence of the input character **82**. The display characters **81** in the input character string display area **30** shown in FIG. **9**A are the replacement characters **83** for replacing the input characters **82**, except for the immediately previous input character **84**. In order to confirm the input characters **82** that are replaced into the replacement characters **83**, the control unit **10** may display a password display button **74** on the display unit **15**, and may display all the input characters **82** included in the character string **80** in the input character string display area **30** when receiving an operation on the password display button **74**.

(97) As shown in FIG. **9**B, the character string **80** may be the SSID. The SSID is an example of the identifier. FIG. **9**B schematically shows an SSID input screen displayed on the display unit **15**. The control unit **10** repeatedly receives, from the character input area **20**, the operation for inputting the input character **82**, and displays the SSID that is the character string **80** in the input character string display area **30**. The example shown in FIG. **9**B is suitable when the SSID acquisition function of the OS does not work.

(98) After the process of S**206** shown in FIG. **8**, the control unit **10** displays the immediately

previous input character **84** in the immediately previous input character display area **40** in a manner of being larger than the display character **81** in the input character string display area **30** (S**208**). The immediately previous input character **84** in the immediately previous input character display area **40** is larger than the character in the character input area **20**.

(99) Here, as shown in FIGS. **2** and **9**B, the control unit **10** may change the typeface depending on whether the immediately previous input character **84** displayed in the immediately previous input character display area **40** is the uppercase character of the alphabet or the lowercase character of the alphabet. Accordingly, the user can easily determine whether the immediately previous input character **84** is the uppercase character of the alphabet or the lowercase character of the alphabet.

(100) Further, as shown in FIG. **9**C, the control unit **10** may change the character size depending on whether the immediately previous input character **84** displayed in the immediately previous input character display area **40** is the uppercase character of the alphabet or the lowercase character of the alphabet. FIG. **9**C schematically shows display of the uppercase character of the alphabet and display of the lowercase character of the alphabet in the immediately previous input character display area **40**. It is shown that in the immediately previous input character display area **40** shown in FIG. **9**C, the immediately previous input character **84** is displayed with a relatively small character size when the immediately previous input character **84** is the lowercase character of the alphabet, and the immediately previous input character **84** is displayed with a relatively large character size when the immediately previous input character **84** is the uppercase character of the alphabet. As shown in FIG. **9**C, the typeface of the immediately previous input character **84** when the immediately previous input character **84** is the uppercase character of the alphabet is different from that when the immediately previous input character **84** is the lowercase character of the alphabet. However, the typeface of the immediately previous input character **84** when the immediately previous input character **84** is the uppercase character of the alphabet may be the same as that when the immediately previous input character **84** is the lowercase character of the alphabet.

(101) Further, as shown in FIG. **10**, the control unit **10** may display, on the display unit **15**, the animation display **25** in which the input character **82** input in response to the reception of the operation floats from the character input area **20** and moves to the immediately previous input character display area **40**. FIG. **10** schematically shows the animation display **25** of the input character **82**. The input character **82** may move continuously or may move in a stepwise manner toward the immediately previous input character display area **40** after floating from the character input area **20**. When the animation display **25** in which the input character **82** floats from the character input area **20** and moves to the immediately previous input character display area **40** is executed, it is easier to confirm the immediately previous input character.

(102) After S**208** shown in FIG. **8**, the control unit **10** acquires the input character confirmation information IN**1** corresponding to the input character **82** from the database DB**1** including the input character confirmation information IN**1** (S**210**). When the database DB**1** is stored in the storage unit **14**, the control unit **10** can acquire the input character confirmation information IN**1** corresponding to the input character **82** from the database DB**1** stored in the storage unit **14**. When the database DB**1** is stored in the server **300**, the control unit **10** transmits an acquisition request for the input character confirmation information IN**1** corresponding to the input character **82** to the server **300**. When receiving the acquisition request, the server **300** reads the input character confirmation information IN**1** corresponding to the input character **82** from the database DB**1**, and transmits the input character confirmation information IN**1** to the display device **1**. The display device **1** can acquire the input character confirmation information IN**1** corresponding to the input character **82** by receiving the input character confirmation information IN**1** from the server **300**.

(103) Here, as shown in FIG. **4**, when the input character confirmation information IN**1** is divided into the plurality of groups of language-specific information IN**4**, the control unit **10** acquires, from the database DB**1**, the language-specific information IN**4** corresponding to the language setting of the display device **1** among the input character confirmation information IN**1** corresponding to the

input character **82**, according to the setting information stored in the storage unit **14**. When the input character confirmation information IN**1**, which is the language-specific information IN**4** corresponding to the language setting, is divided into a plurality of groups of confirmation information IN**2**, the control unit **10** acquires, from the database DB**1**, the confirmation information IN**2** selected among the input character confirmation information IN**1** corresponding to the input character **82**, according to the group setting stored in the storage unit **14**. When the edited input character confirmation information IN**1** is stored in the database DB**1**, the control unit **10** acquires the edited input character confirmation information IN**1** corresponding to the input character **82** from the database DB**1**. When the edited input character confirmation information IN**1** is stored in the storage unit **14** separately from the database DB**1**, the control unit **10** acquires the edited input character confirmation information IN**1** from the storage unit **14** when the input character confirmation information IN**1** corresponding to the input character **82** is edited.
(104) For example, it is assumed that the language setting of the display device **1** is English, and the selection area **60***b* of “emphasis on easy-to-understand word” is selected from the confirmation information selection area **60** shown in FIG. **6**. In this case, the control unit **10** acquires confirmation information corresponding to the input character **82** from the confirmation information IN**2***b* with emphasis on the easy-to-understand word among the input character confirmation information IN**1** shown in FIG. **3**. As shown in FIG. **2**, when the immediately previous input character **84** is “A”, the control unit **10** acquires “APPLE” that is the confirmation information IN**2***b* with emphasis on the easy-to-understand word among the input character confirmation information IN**1** corresponding to the input character “A”. As shown in FIG. **9**B, when the immediately previous input character **84** is “a”, the control unit **10** acquires “apple” that is the confirmation information IN**2***b* with emphasis on the easy-to-understand word among the input character confirmation information IN**1** corresponding to the input character “a”.
(105) After S**210** shown in FIG. **8**, the control unit **10** displays the input character confirmation information IN**1** acquired from the database DB**1** following the immediately previous input character **84** in the immediately previous input character display area **40** (S**212**), and returns the process to S**204**. As shown in FIGS. **2** and **9**B, the input character confirmation information IN**1** displayed in the immediately previous input character display area **40** has the same character size as the immediately previous input character **84** displayed in the immediately previous input character display area **40**. Since the input character confirmation information IN**1** is displayed in the large size, the user can easily confirm the immediately previous input character. In addition, the control unit **10** displays the input character confirmation information IN**1**, corresponding to the immediately previous input character **84**, in parentheses in the immediately previous input character display area **40**. Since the user can view the input character confirmation information IN**1** displayed in the parentheses according to the input character input in response to the immediately previous operation, the user can more easily confirm the immediately previous input character.
(106) Here, as shown in FIG. **9**C and the like, the control unit **10** may display the input character confirmation information IN**1** in the immediately previous input character display area **40** such that at least one of the typeface and the character size is changed depending on whether the input character **82** is the uppercase character of the alphabet or the lowercase character of the alphabet. When the input character confirmation information IN**1** is displayed such that at least one of the typeface and the character size when the input character **82** is the uppercase character of the alphabet is different from that when the input character **82** is the lowercase character of the alphabet, the immediately previous input character can be more easily confirmed.
(107) As shown in FIG. **4**, when the input character confirmation information IN**1** is divided into a plurality of groups of language-specific information IN**4**, the language-specific information IN**4** corresponding to the language setting of the display device **1** among the input character confirmation information IN**1** corresponding to the input character **82** is displayed in the immediately previous input character display area **40**. Therefore, the user can more easily confirm

the immediately previous input character.

(108) When the input character confirmation information IN**1**, which is the language-specific information IN**4** corresponding to the language setting, is divided into a plurality of groups of confirmation information IN**2**, the confirmation information IN**2** corresponding to the group setting among the input character confirmation information IN**1** corresponding to the input character **82** is displayed in the immediately previous input character display area **40**. Therefore, the user can more easily confirm the immediately previous input character.

(109) For example, when the language setting of the display device **1** is English and the selection region **60***b* of the "emphasis on easy-to-understand word" is selected from the confirmation information selection area **60** shown in FIG. **6**, the confirmation information IN**2***b* with emphasis on the easy-to-understand word among the input character confirmation information IN**1** shown in FIG. **3** is displayed in the immediately previous input character display area **40**. As shown in FIG. **2**, when the immediately previous input character **84** is "A", the confirmation information "APPLE" corresponding to the input character "A" among the confirmation information IN**2***b* with emphasis on the easy-to-understand word is displayed in the immediately previous input character display area **40**. As shown in FIG. **9**B, when the immediately previous input character **84** is "a", the confirmation information "apple" corresponding to the input character "a" among the confirmation information IN**2***b* with emphasis on the easy-to-understand word is displayed in the immediately previous input character display area **40**.

(110) When receiving the editing of the input character confirmation information IN**1** corresponding to the input character **82**, the control unit **10** displays the input character confirmation information IN**1**, which corresponds to the input character **82** and is obtained in response to reception of the editing, in the immediately previous input character display area **40**. Since the customized input character confirmation information IN**1** is displayed, the user can more easily confirm the immediately previous input character.

(111) By repeating the processes of S**204** to S**212** shown in FIG. **8**, the control unit **10** repeatedly receives, from the character input area **20**, the operation for inputting the input character **82**, arranges the input character **82** or the replacement character **83** in order in the input character string display area **30**, and displays the immediately previous input character **84** and the input character confirmation information IN**1** corresponding to the immediately previous input character **84** in the immediately previous input character display area **40**.

(112) In a determination process of S**204** shown in FIG. **8**, when the backspace key **21***e* shown in FIG. **2** is operated, the control unit **10** deletes the character at the tail of the character string **80** displayed in the input character string display area **30**, and returns the process to S**204**. In the determination process of S**204** shown in FIG. **8**, when the keyboard switching button **71** shown in FIG. **2** is operated, the character input area **20** shown in FIG. **2** and the software keyboard having the QWERTY layout are switched, and the process returns to S**204**. In the determination process of S**204** shown in FIG. **8**, when the cancel button **73** shown in FIG. **2** is operated, the control unit **10** discards the character string **80** displayed in the input character string display area **30** and ends the character string input reception process.

(113) When the enter key **21***f* or the OK button **72** is operated in the determination process of S**204** shown in FIG. **8**, the control unit **10** determines the character string **80** displayed in the input character string display area **30**, transmits the character string **80** to the next process (S**214**), and ends the character string input reception process.

(114) As described above, when the user operates the character input area **20**, the input character **82** or the replacement character **83** is arranged in order in the input character string display area **30**, and the input character confirmation information IN**1** corresponding to the input character **82** is displayed in the immediately previous input character display area **40**. The user can confirm whether the input character **82** is correct by viewing the input character confirmation information IN**1** corresponding to the input character **82**. As described above, in this specific example, since

what the immediately previous input character is can be easily confirmed, the erroneous input of important registration information such as the password, the identifier, and the like can be reduced, failure of setting such as wireless connection can be reduced.

4. Modifications

(115) Various modifications of the present disclosure are conceivable.

(116) For example, when the replacement character **83** is arranged in the input character string display area **30** as shown in FIG. **9**A, the immediately previous input character **84** may be replaced with the replacement character **83**.

(117) The control unit **10** may receive setting as to whether to arrange the input character **82** or the replacement character **83** in the input character string display area **30**. In addition, the control unit **10** may arrange the input character **82** in the input character string display area **30** when the setting is to arrange the input character **82**, and arrange the replacement character **83** in the input character string display area **30** when the setting is to arrange the replacement characters **83**.

(118) The above process can be appropriately changed, for example, an order is changed. For example, in the character string input reception process shown in FIG. **8**, the process of S**206** for arranging the input character **82** in the input character string display area **30** can be executed after any of the processes of S**208**, S**210**, and S**212**.

(119) As shown in FIG. **11**, the control unit **10** may set a group for displaying the input character confirmation information IN**1** according to the country or the region where the display device **1** is present. FIG. **11** shows another example of the group setting process executed by the control unit **10**.

(120) When the group setting process is started, the control unit **10** acquires information of the country or the region where the display device **1** is present (S**122**). The OS of the display device **1** causes the display device **1** to implement a function of acquiring information of the country or the region where the display device **1** is present and storing the information in the storage unit **14**, and a function of reading the information of the country or the region from the storage unit **14** in response to a country or region setting acquisition request from an application program and transmitting the information to the application program. In addition, the OS of the display device **1** may cause the display device **1** to implement a function of receiving setting of the country or the region and storing the setting in the storage unit **14**, and a function of reading the setting of the country or the region from the storage unit **14** in response to the country or region setting acquisition request from the application program and transmitting the information of the country or the region to the application program. The control unit **10** can execute a process of acquiring the information of the country or the region from the OS by issuing the country or region setting acquisition request to the OS. After the process of S**122**, the control unit **10** executes a process of setting, based on the acquired information of the country or the region, the country-specific or region-specific information IN**3** corresponding to the country or the region as the input character confirmation information IN**1** to be used, for example, a process of storing, in the storage unit **14**, setting information indicating the country-specific or region-specific information IN**3** corresponding to the country or the region (S**124**).

(121) The input character confirmation information IN**1**, which is the country-specific or region-specific information IN**3** corresponding to the country or the region, may be divided into a plurality of groups of confirmation information IN**2** as shown in FIG. **3**. When the input character confirmation information IN**1** is divided into groups, the control unit **10** displays the confirmation information selection area **60** as shown in FIG. **6** on the display unit **15** (S**126**). After the confirmation information selection area **60** is displayed, the control unit **10** stores, in the storage unit **14**, group setting indicating confirmation information corresponding to an area selected from the selection areas **60***a*, **60***b*, and **60***c* (S**128**), and ends the group setting process.

(122) When the group setting process shown in FIG. **11** is executed, in S**210** shown in FIG. **8**, the control unit **10** acquires, from the database DB**1**, the confirmation information IN**2** selected among

the country-specific or region-specific information IN**3** corresponding to the input character **82**, according to the group setting stored in the storage unit **14**. Accordingly, in S**212** shown in FIG. **8**, the country-specific or region-specific information IN**3** corresponding to the country or the region where the display device **1** is present among the input character confirmation information IN**1** corresponding to the input character **82** is displayed in the immediately previous input character display area **40**. Therefore, the user can more easily confirm the immediately previous input character.

(123) As shown in FIG. **12**, the present technique can also be applied when the display unit **15** does not display the immediately previous input character display area **40**. FIG. **12** schematically shows another password input screen displayed on the display unit **15** of the display device **1**. The password input screen includes the character input area **20**, the input character string display area **30**, the keyboard switching button **71**, the OK button **72**, the cancel button **73**, and the password display button **74**. As shown in FIG. **12**, the plurality of soft keys **21** can have various color schemes, such as not being color-coded. On the password input screen shown in FIG. **12**, the immediately previous input character display area **40** is not displayed, and the input character confirmation information IN**1** is displayed in a balloon according to the immediately previous input character **84** arranged at the tail of the character string **80** displayed in the input character string display area **30**. The control unit **10** sequentially arranges the replacement characters **83** in the input character string display area **30** except for the immediately previous input character **84**, acquires the input character confirmation information IN**1** corresponding to the immediately previous input character **84** from the database DB**1**, and displays the acquired input character confirmation information IN**1** in the balloon described above. In the example, since what the immediately previous input character is can also be easily confirmed, the erroneous input can be reduced.

(124) Although not shown, the input character confirmation information IN**1** may include only the phonetic code of the character **90**, or may include only the word having the character **90** as the initial character. Of course, the input character confirmation information IN**1** may not be divided into a plurality of groups of confirmation information IN**2**. The confirmation information selection area **60** shown in FIG. **6** may not be displayed on the display unit **15**. The group setting process shown in FIGS. **5** and **11** may not be executed.

5. Conclusion

(125) As described above, according to the present disclosure, a technique for reducing the erroneous input can be provided based on various aspects. Of course, the above basic functions and effects can be obtained even with a technique which does not include constituent requirements according to the dependent claims but includes only the constituent requirements according to the independent claims.

(126) In addition, it is also possible to implement a configuration in which configurations disclosed in the above examples are replaced with each other or a combination thereof is changed, a configuration in which configurations disclosed in known techniques and the above examples are replaced with each other or a combination thereof is changed, and the like. The present disclosure also includes these configurations and the like.

Claims

1. A display device, comprising: a display unit including a touch panel; and a control unit configured to: display, on the display unit, a character input area in which a soft key is arranged; repeatedly receive, from the character input area, an operation for inputting an input character; display, on the display unit, an input character string display area in which a character string that is a sequence of the input character is displayed, wherein the character string is one of a password and an identifier; arrange in order, in the input character string display area, one of the input character and a replacement character for replacing the input character; acquire from a database including

input character confirmation information for confirming a character, when receiving the operation for inputting the input character from the character input area, the input character confirmation information corresponding to the input character, wherein the input character confirmation information includes at least one of a phonetic code of the character, a word having the character as an initial character, and a pronunciation of the character; and display the acquired input character confirmation information on the display unit.

2. The display device according to claim 1, wherein the control unit is configured to display, on the display unit, an immediately previous input character display area, in which an immediately previous input character input in response to immediately previous reception of the operation is displayed, separately from the input character string display area.

3. The display device according to claim 2, wherein the control unit is configured to display the immediately previous input character in the immediately previous input character display area in a manner of being larger than a display character in the input character string display area.

4. The display device according to claim 2, wherein the control unit is configured to display the input character confirmation information in parentheses in the immediately previous input character display area.

5. The display device according to claim 2, wherein the control unit is configured to display, on the display unit, the input character input in response to the reception of the operation, such that the input character floats from the character input area and moves to the immediately previous input character display area.

6. The display device according to claim 1, wherein the control unit is configured to display the input character confirmation information on the display unit such that at least one of a typeface and a character size is changed depending on whether the input character is an uppercase character of an alphabet or is a lowercase character of the alphabet.

7. The display device according to claim 1, wherein the control unit is configured to display, on the display unit, an editing area for receiving editing of the input character confirmation information to be displayed on the display unit, and display on the display unit, when receiving the editing of the input character confirmation information from the editing area, the input character confirmation information that corresponds to the input character and is obtained in response to the reception of the editing.

8. The display device according to claim 1, wherein the input character confirmation information is divided into a plurality of groups of confirmation information, and the control unit is configured to display, on the display unit, a confirmation information selection area for receiving selection of confirmation information to be displayed on the display unit from the plurality of groups of confirmation information, acquire from the database, when receiving the selection of the confirmation information from the confirmation information selection area, the confirmation information selected among the input character confirmation information corresponding to the input character, and display the acquired confirmation information on the display unit.

9. The display device according to claim 1, wherein the input character confirmation information is divided into a plurality of groups of country-specific or region-specific information according to countries or regions, and the control unit is configured to acquire, from the database, country-specific or region-specific information corresponding to a country or a region where the display device is present among the input character confirmation information corresponding to the input character, and display the acquired country-specific or region-specific information on the display unit.

10. The display device according to claim 1, wherein the input character confirmation information is divided into a plurality of groups of language-specific information according to languages, and the control unit is configured to acquire, from the database, language-specific information corresponding to a language set in the display device among the input character confirmation information corresponding to the input character, and display the acquired language-specific

information on the display unit.

11. A display method for displaying, on a display unit including a touch panel, a character input area in which a soft key is arranged; repeatedly receiving, from the character input area, an operation for inputting an input character; displaying, on the display unit, an input character string display area in which a character string that is a sequence of the input character is displayed, wherein the character string is one of a password and an identifier; and arranging in order, in the input character string display area, one of the input character and a replacement character for replacing the input character, the display method comprising: acquiring from a database including input character confirmation information for confirming a character, when the operation for inputting the input character is received from the character input area, the input character confirmation information corresponding to the input character, wherein the input character confirmation information includes at least one of a phonetic code of the character, a word having the character as an initial character, and a pronunciation of the character; and displaying the acquired input character confirmation information on the display unit.

12. A non-transitory computer-readable storage medium storing a display program for displaying, on a display unit including a touch panel, a character input area in which a soft key is arranged, repeatedly receiving, from the character input area, an operation for inputting an input character, displaying, on the display unit, an input character string display area in which a character string that is a sequence of the input character is displayed, wherein the character string is one of a password and an identifier, and arranging in order, in the input character string display area, one of the input character and a replacement character for replacing the input character, the display program causing a computer to implement: an input character confirmation information acquisition function of acquiring from a database including input character confirmation information for confirming a character, when the operation for inputting the input character is received from the character input area, the input character confirmation information corresponding to the input character, wherein the input character confirmation information includes at least one of a phonetic code of the character, a word having the character as an initial character, and a pronunciation of the character; and an input character confirmation information display function of displaying the acquired input character confirmation information on the display unit.